

Project Report

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1. Project Overview

MaritimeMonitor is a real-time maritime intelligence dashboard built as a React 18 + TypeScript single-page application. It renders a 3D interactive globe displaying live AIS vessel tracking data alongside port intelligence, chokepoint risk, and search functionality.

Layer	Technology
Frontend framework	React 18 + TypeScript + Vite
3D globe	react-globe.gl + Three.js
State management	Zustand 4 (Map<mmsi, Vessel>)
Styling	Tailwind CSS
AIS relay server	Node.js + ws (WebSocket library)
Primary AIS source	AISStream.io — free WebSocket
Secondary AIS	DataDocked — satellite REST API

2. What Was Built

2.1 3D Globe

- ✓ NASA Blue Marble texture (earth-blue-marble.jpg) for realistic Earth appearance
- ✓ Topology bump map for surface relief
- ✓ Night-sky starfield background
- ✓ Atmospheric limb glow (blue)
- ✓ Auto-rotation at 0.35 rpm with damped orbit controls
- ✓ Zoom limits: 105 (max zoom-in) → 700 (max zoom-out)
- ✓ flyTo animation registered in Zustand so search can pan the camera

2.2 AIS Data Pipeline

- ✓ AIS relay server (server/relay.js) — persistent WebSocket to AISStream.io
- ✓ Relay accumulates vessels in-memory 24/7; clients get instant snapshot on connect
- ✓ Vessel name cleaning: strips AIS padding characters (@@@ and ____)
- ✓ Batch store updates every 2 s — reduced React re-renders from ~300/s to 0.5/s
- ✓ DataDocked satellite AIS service — polls key shipping hubs via REST
- ✓ Prune stale vessels (>10 min without update) every 60 s

2.3 Multi-provider Architecture

The app checks for the relay first, falls back to direct AISStream if the relay is not running:

Priority	Provider	Type	Cost
1 (preferred)	Local relay → AISStream.io	Terrestrial WebSocket	Free
2 (parallel)	DataDocked	Satellite REST	Credits

2.4 Search

- ✓ Word-by-word normalised matching (handles AIS padding, underscores, extra spaces)
- ✓ Searches: name, MMSI, flag, vessel type, callsign, IMO, destination
- ✓ Keyboard navigation: arrow keys, Enter to select, Escape to dismiss
- ✓ On select: camera flies to vessel position then opens info panel
- ✓ Results memoised with useMemo — no re-filter on unrelated renders

2.5 Performance Optimisations

Problem	Fix	Impact
300 store updates/s	Batch flush every 2 s	Re-renders: 300/s → 0.5/s
18k points recomputed	useMemo on vesselPoints	Only recalculates on flush
180k polygon segments	pointResolution 10 → 4, cap 12k	60 % less GPU geometry
Search filters 18k	useMemo on results	Only recalculates on type

3. Bugs Fixed

Bug	Root Cause	Fix
Globe showed hollow earth	Custom ocean material with transparent land	Replaced with NASA Blue Marble globeImageUrl
Status bar showed 0 vessels	loadMockVessels() only called when no API key	Always seed mocks first, layer live AIS on top
'African Heron' not found	AIS names arrive as 'AFRICAN_HERON@ @ @'	norm() strips padding; wordMatch() splits query
DataDocked silent failure	Wrong endpoint (POST bounding box vs GET radius)	Rewrote with correct GET ?lat&lon&circle_radius=1000
DataDocked auth rejected	Used Authorization: Bearer — API uses x-api-key	Switched to x-api-key header
TypeScript errors on build	Missing @types/three, lineWidthMinPixels callback	Installed types; changed callback to static value

4. Current Problems

4.1 Open Ocean Coverage Gap ■ Major

All 18,000+ vessels are located along coastlines. The open Atlantic, Pacific, and Indian oceans appear empty. This is a fundamental physical limitation of terrestrial AIS — ground receivers only reach ~50 km from shore.

- ✗ Ships transiting the open ocean are completely invisible
- ■ DataDocked satellite AIS is integrated and correct but the free trial credits (10) are exhausted
- ■ Global Fishing Watch API investigated — returns 96-hour-old heatmap data, not real-time positions
- ■ Spire (acquired by Kpler) no longer has a free developer tier

Resolution: Top up DataDocked credits. The service is already wired in correctly (GET /get-vessels-by-area, x-api-key header, 100 km radius polling of 10 key shipping hubs). Adding credits will immediately populate open-ocean positions.

4.2 DataDocked Credits Exhausted

■ ❌ Account balance: 10 credits — exactly enough for 1 area call (costs 10 credits each)

■ ■ On startup the service queries Singapore Strait (world's busiest hub) then stops

Resolution: Purchase credits at datadocked.com. The polling rotation covers 10 hubs: Singapore, English Channel, Malacca, Suez, Hormuz, Gulf of Aden, Rotterdam, New York, Shanghai, Gulf of Mexico — one hub per 10 minutes.

4.3 Relay Server Must Be Started Manually

■ ■ npm run start launches both relay and Vite — but the relay is local only

■ ■ If the relay is not running, the app falls back to direct AISStream (slower initial load)

■ ■ No persistent hosting — relay loses its vessel cache if restarted

Resolution: Deploy relay to a free cloud host (Railway, Fly.io, Render). The relay.js file is self-contained and ready to deploy.

4.4 Vessel Detail Panel — Incomplete Data

■ ■ AISStream PositionReport messages carry no vessel type — typeCode defaults to 0 (unknown)

■ ■ ShipStaticData messages fill in type/name/callsign but arrive less frequently

■ ■ Many vessels show 'Unknown' type and no flag in the detail panel

Resolution: Merge PositionReport and ShipStaticData in the relay (partially done — relay merges with existing vessel using spread). Could be improved by caching static data longer than 10-minute prune window.

5. Key Files

File	Purpose
server/relay.js	AIS relay — persistent WS to AISStream, HTTP snapshot + WS broadcast
src/hooks/useAIS.ts	Orchestrates relay / AISStream / DataDocked — picks best source
src/services/relayService.ts	Browser WS client for relay — buffers updates, flushes every 2 s
src/services/aisStream.ts	Direct AISStream WebSocket fallback
src/services/dataDockedService.ts	Satellite AIS — GET radius queries, 10 shipping hubs
src/store/maritimeStore.ts	Zustand store — vessels Map, batchUpsertVessels, flyToFn
src/components/Globe/GlobeContainer.tsx	react-globe.gl — memoised points, auto-rotate, flyTo
src/components/UI/SearchBar.tsx	Normalised word-match search across all vessel fields

6. Running the App

Start everything (relay + Vite dev server):

```
npm run start
```

Relay only:

```
npm run relay
```

Vite only (relay fallback to direct AISStream):

```
npm run dev
```

Type-check:

```
npx tsc --noEmit
```